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CIVL4340
Wind Engineering

Week 2: Wind climate analysis

Overview

- **Aims of a wind climate analysis**
- **Basic wind statistics**
  - Parent wind distribution
  - Extreme value distributions
- **Types of extreme value analyses**
- **Reading:**
  - Holmes, Chapter 2
  - Simiu and Yeo, Chapter 3 and parts of Appendix A

Wind climate analysis

- **Design wind speeds**

Regional wind speed (m/s)	Regions				
	Non-cyclonic		Cyclonic		
	A (0 to 5)	B1	B2	C (maximum)	D (maximum)
V_1	30	26	26	23	23
V_5	32	28	28	33	35
V_{10}	34	33	33	39	43
V_{20}	37	38	38	45	51
V_{25}	37	39	39	47	53
V_{50}	39	44	44	52	60
V_{100}	41	48	48	56	66
V_{200}	43	52	52	61	72
V_{250}	43	53	53	62	74
V_{500}	45	57	57	66	80
V_{1000}	46	60	60	70	85
V_{2000}	48	63	63	73	90
V_{2500}	48	64	64	74	91
V_{5000}	50	67	67	78	95
V_{10000}	51	69	69	81	99
V_R ($R \geq 5$ years)	$67–41R^{-0.1}$	$106–92R^{-0.1}$	$106–92R^{-0.1}$	$122–104R^{-0.1}$	$156–142R^{-0.1}$

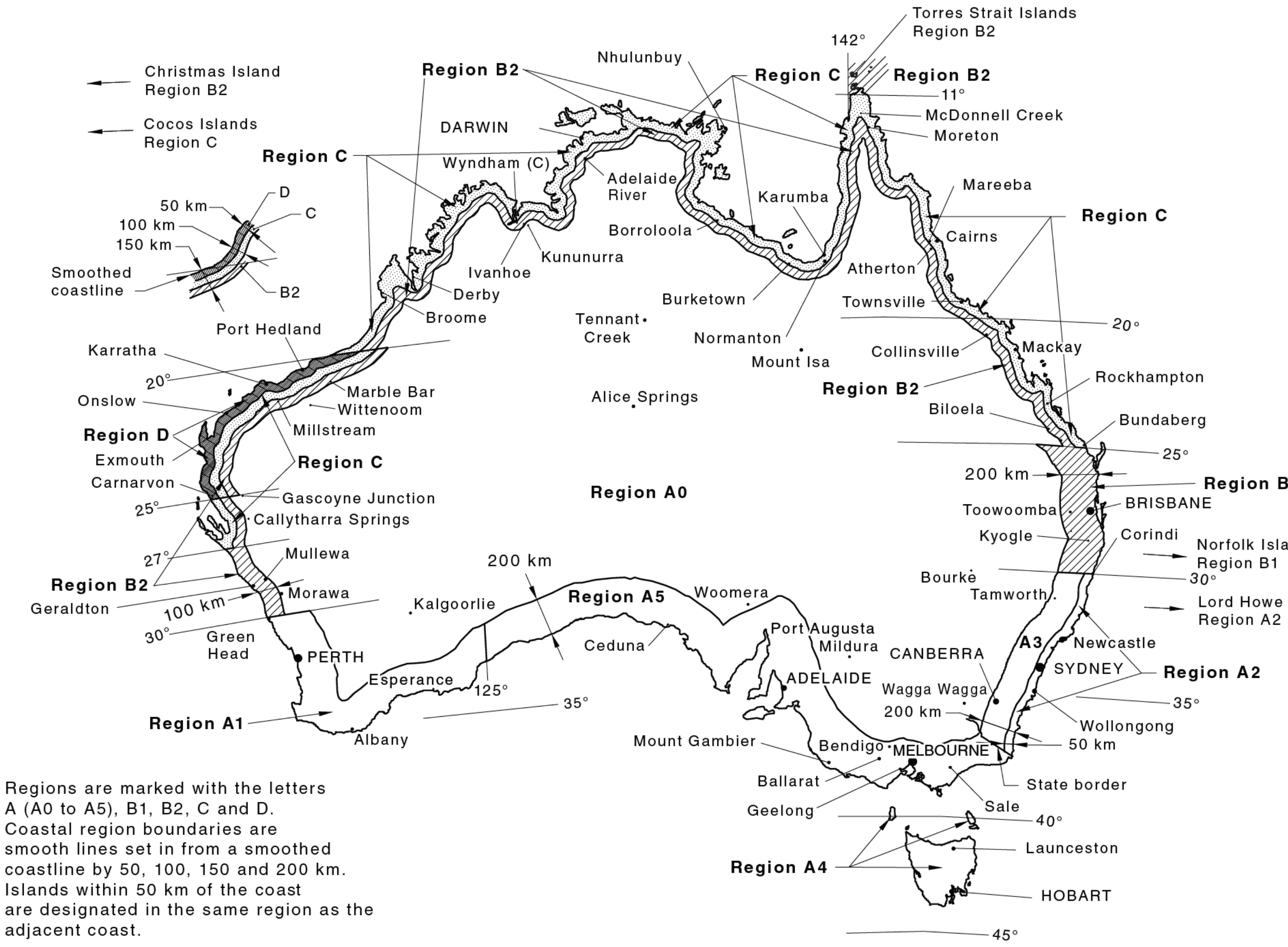
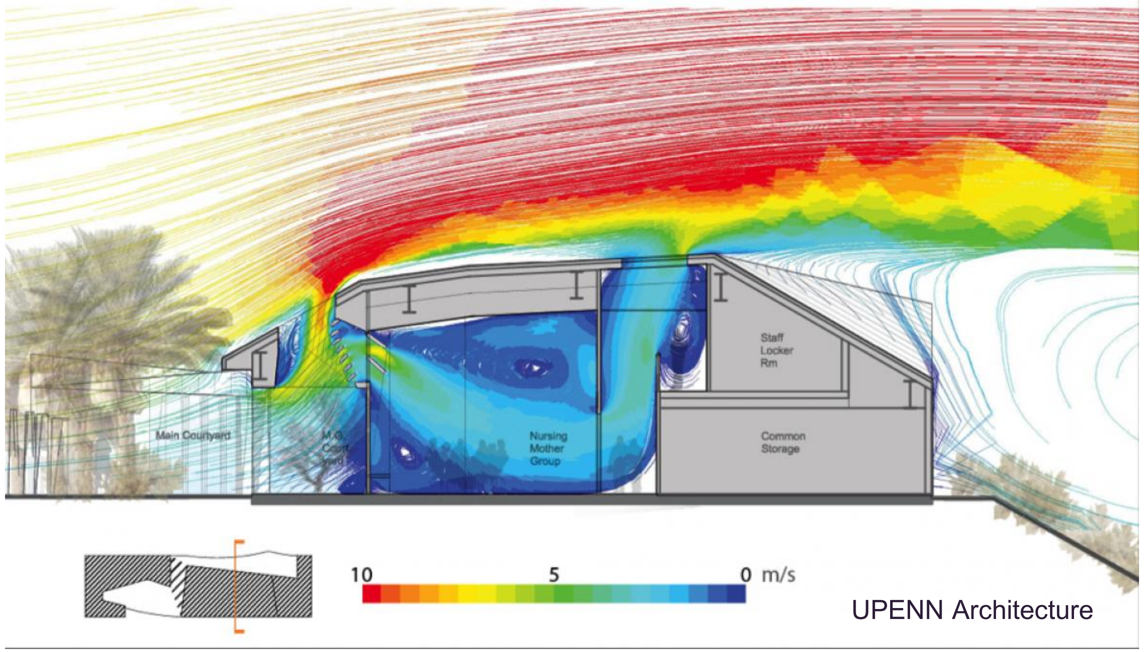


Figure 3.1(A) — Wind regions — Australia

Wind climate analysis

- **Fatigue analysis**
- **Pedestrian safety/comfort**
- **Building physics**

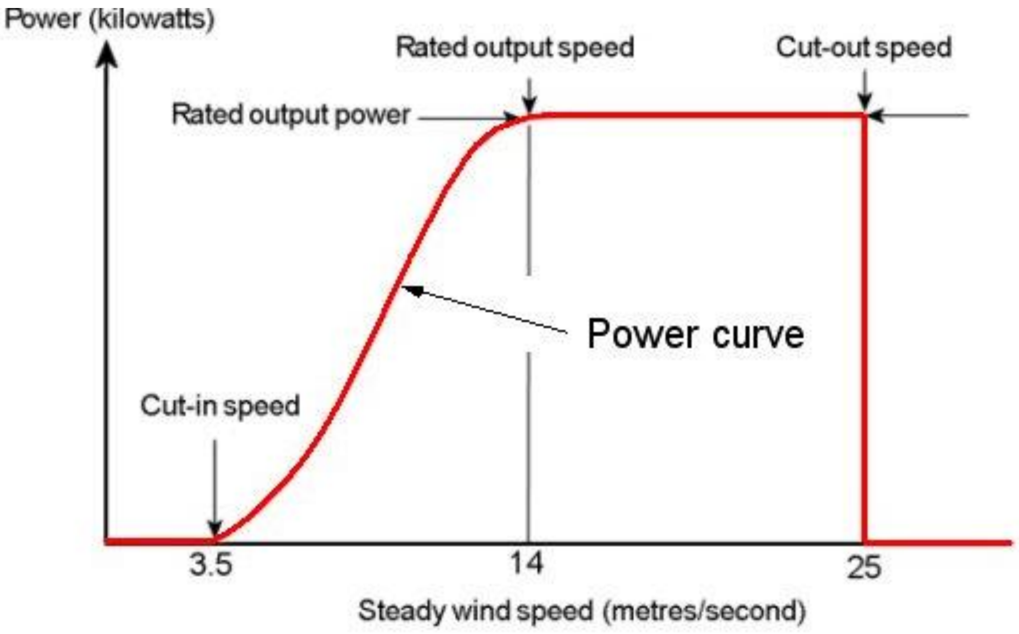
Holmes (2002)



UPENN Architecture

Wind climate analysis

- **Wind power estimation**
- **Energy (infrastructure) security**

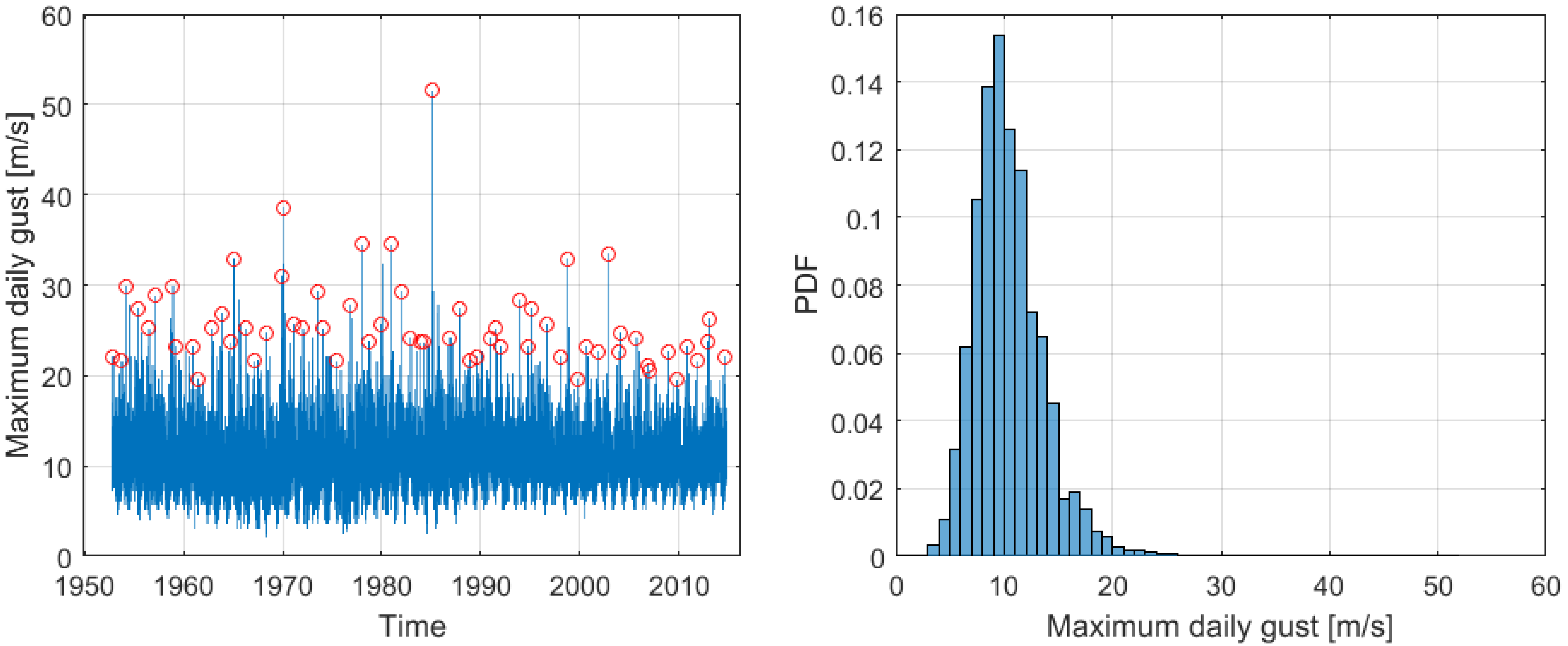


Typical wind turbine power output with steady wind speed.

WINDPOWER program

Wind statistics

Parent distribution



Brisbane Airport (daily maximum gust)

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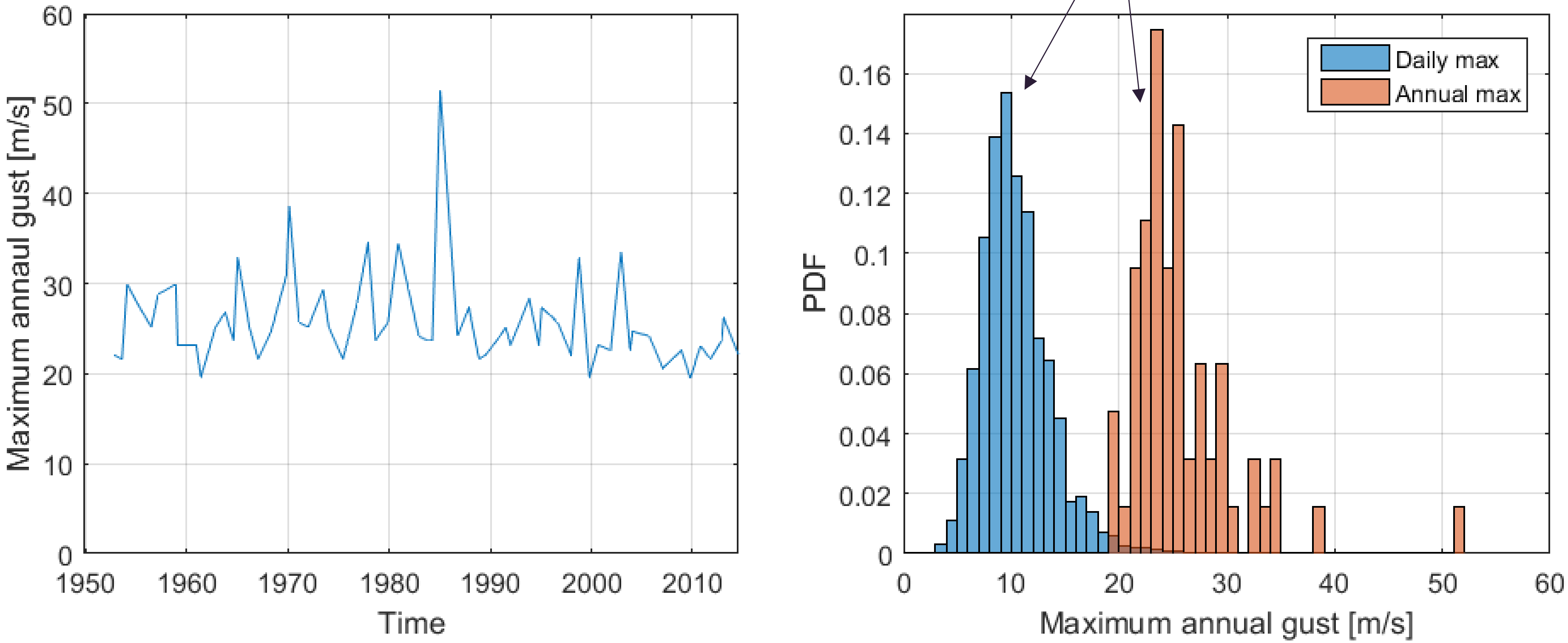
Wind statistics

Extrema

Similar, but different distributions



Brisbane Airport (annual maximum gust)

Wind statistics

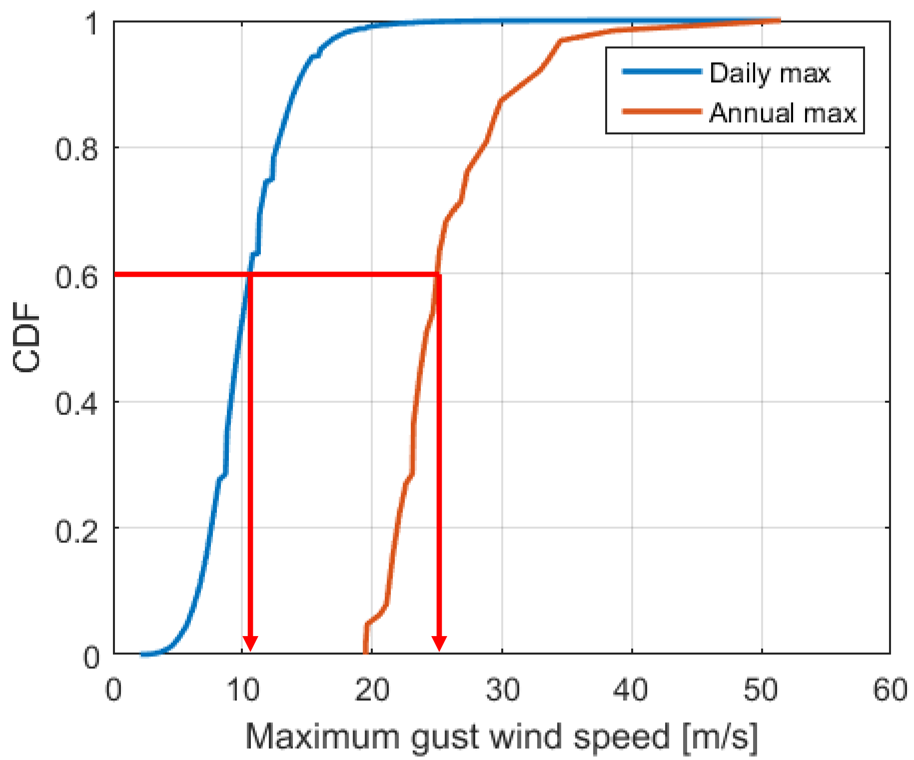
Cumulative probability

Prob. a value is NOT exceeded

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx$$

$$G(x) = P(X \geq x) = 1 - F(x)$$

Prob. a value IS exceeded



Wind statistics

- Often don't have sufficiently long data records to estimate design wind speed (e.g. RP or ARI = 500 yrs)
- We therefore want to fit known distributions to data so we can extrapolate.
- First assumed a normal distribution (based on parent) was suitable – this was shown to be inadequate for extreme geophysical phenomena.
- Distributions just for the extrema developed.
- Most geophysical phenomena can be treated similarly.

Extreme value analysis

Generalised Extreme Value distribution

- Three asymptotic distributions

Position factor

Shape factor

Wind speed

$$F(U) = e^{-\left[1-\frac{k(U-u)}{a}\right]^{1/k}}$$

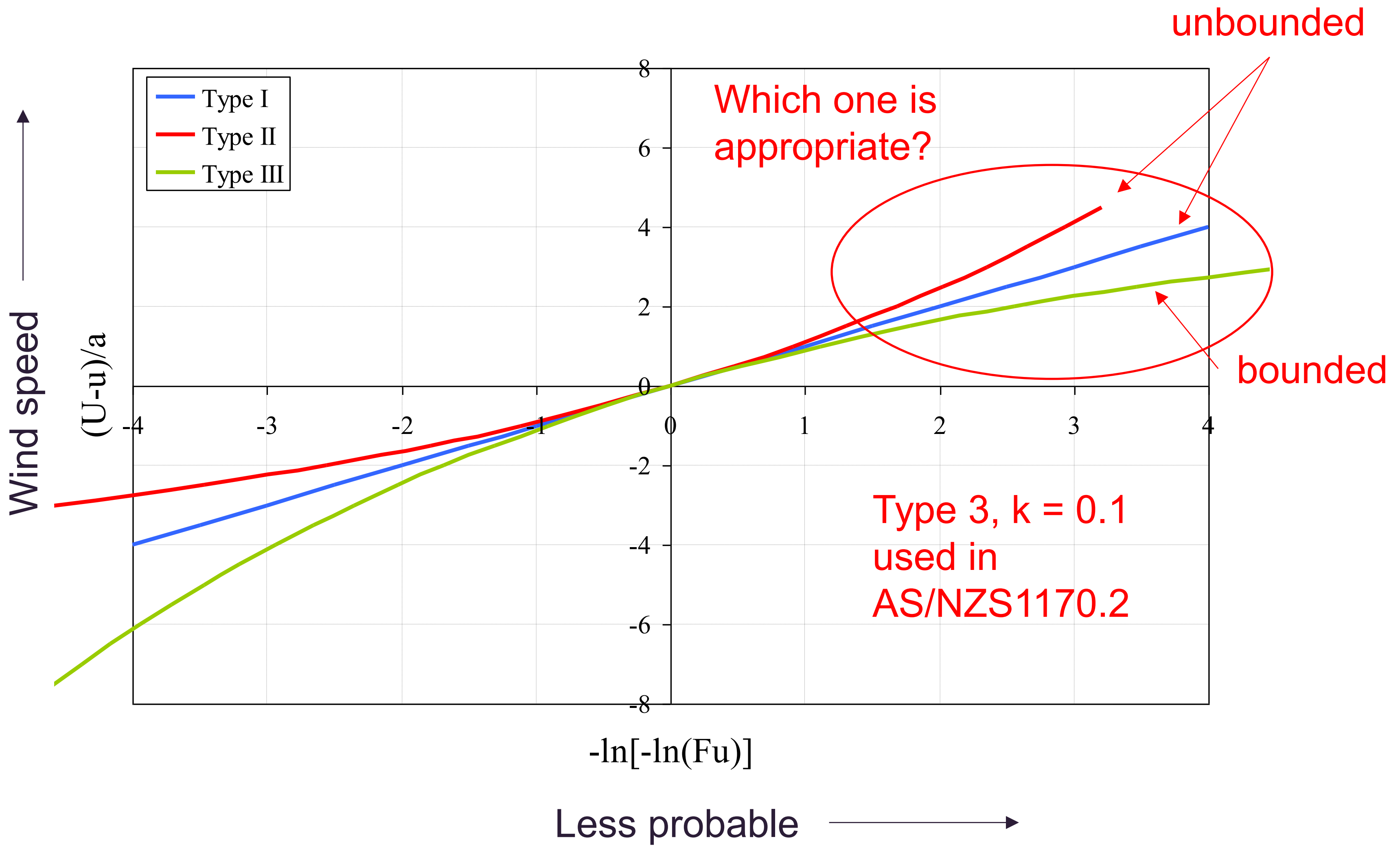
Scale factor

k = 0 (Type 1 - Gumbel)
k < 0 (Type 2)
k > 0 (Type 3)

- For Gumbel distribution (k = 0) this simplifies to,

$$F(U) = e^{-\left[e^{-\left(\frac{U-u}{a}\right)}\right]}$$

Extreme value analysis



Return period

We must now relate the reduced variate to a return period, R (or ARI)

$$R = \frac{1}{\text{Annual probability } \textbf{of exceedance}}$$

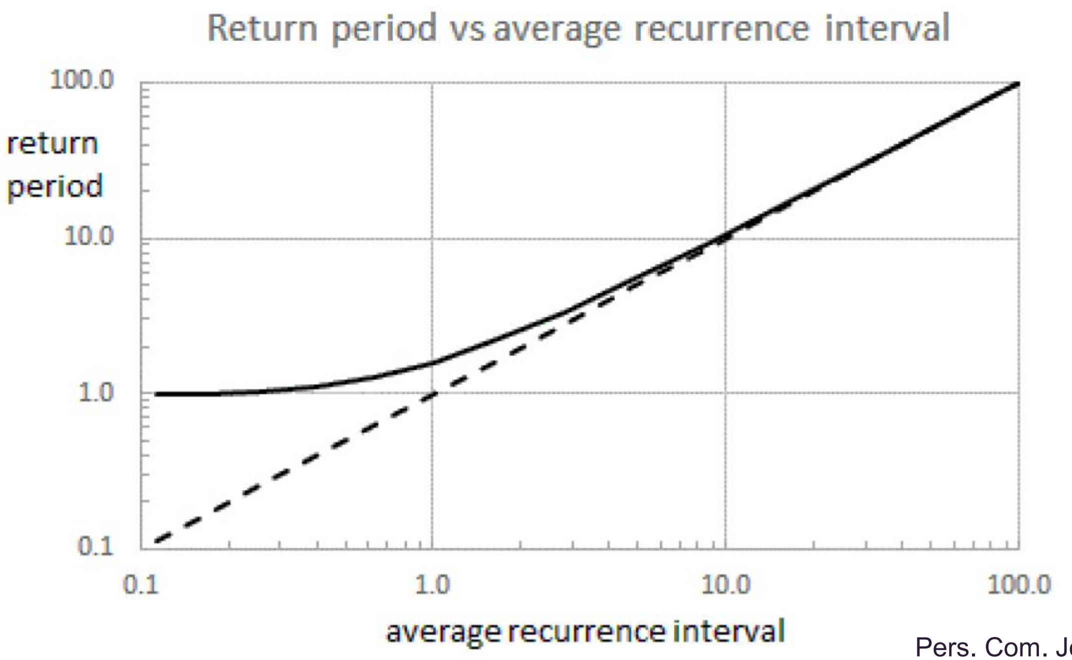
Must have an associated time period

$$R = \frac{1}{G(U)} = \frac{1}{1 - F(U)}$$

NOTE: RETURN PERIOD VS ARI

- Terms, Return Period (R) and Average Recurrence Interval (ARI) are often used interchangeably – but not always correctly.
- ARI becoming more prevalent in Australian Standards.

$$\frac{1}{R} = 1 - e^{\left(-\frac{1}{ARI}\right)}$$



Pers. Com. John Holmes

Probability of exceedance

- Mathematically we can determine the probability wind speed will occur within a given period (e.g. working life of a structure) using the Binomial Distribution.
- This is how we relate R to structural safety.

Number of samples (years)

APE = 1/R

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Probability of exactly *x* occurrences

Binomial coefficient

$$\binom{n}{x} = \frac{n!}{x!\,(n-x)!}$$

Probability of 1 or more exceedances

Probability of no exceedances

$$P(X \geq 1) = 1 - P(X = 0)$$

Probability of exceedance

What does this mean for design wind speeds?

Period of interest

$$P(X \ge 1) = 1 - \cancelto{1}{\binom{n}{0}}\,\cancelto{1}{p^0}(1-p)^{n-0} = 1 - \underbrace{\left(1-\frac{1}{R}\right)}_{\text{P(X=0)}}{}^{n}$$

For a structure that has a design life of 50 years

R = 50yrs

$$P(X \ge 1) = 1 - \left(1 - \frac{1}{50}\right)^{50} = 0.64 \quad [64\%]$$

R = 1000yrs

$$P(X \ge 1) = 1 - \left(1 - \frac{1}{1000}\right)^{50} = 0.049 \quad [5\%]$$

This approach makes assumptions about event occurrence rates but is reasonable for geophysical flows.

Example

Using the Brisbane Airport wind records discussed throughout this lecture,

a) Estimate the 25 year and 500 year return period (R) wind gusts at the site using the Gumbel approach (i.e. GEV-I).

b) Compare these values with those in AS/NZS1170.2 for the region.

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Gumbel approach

$$F(U) = e^{-\left[e^{-\left(\frac{U-u}{a}\right)}\right]}$$

Type 1 GEV

Rearrange for *U* then introduce *R*

$$U = u + a\{-ln[-ln(F(U))]\}$$

$$R = \frac{1}{1 - F(U)}$$

$$U = u + a\left\{-\ln\left[-\ln\left(1 - \frac{1}{R}\right)\right]\right\}$$

Which simplifies to

$$U = u + a\{y\}$$

How to estimate y?
(reduced variate)

Gumbel approach

1. Clean and check data for completeness
2. Extract largest wind speed per year
3. Rank series in ascending order
4. Assign a value of *p* to each series member based on its rank (note that *p* is an approximation of *F*(*U*))
5. Calculate the reduced variate, *y*
6. Plot data and determine u and a.

rank

$$p = \frac{m}{N+1}$$

Total number of data points

$$y = -\ln\left[-\ln(p)\right]$$

NOT THE SAME DATA

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	Q	R	S
	year	gust [m/s]	
	1952	31.4	
	1953	33.4	
	1954	29.8	
	1955	30.3	
	1956	27.8	
	1957	30.3	
	1958	29.3	
	1959	36.5	
	1960	29.3	
	1961	27.3	
	1962	31.9	
	1963	28.8	
	1964	25.2	
	1965	27.3	
	1966	23.7	
	1985	30.3	
	1986	28.3	
	1987	30.3	
	1988	34	
	1989	28.8	
	1990	30.3	
	1991	27.3	
	1992	27.8	
	1993	28.8	
	1994	30.9	
	1995	26.2	
	1996	25.7	
	1997	24.7	
	1998	42.2	

	A	B	C	D	E
1					
2	year	gust [m/s]	rank	p	y
3	1966	23.7	1	0.020833	-1.35856
4	1997	24.7	2	0.041667	-1.15627
5	1964	25.2	3	0.0625	-1.01978
6	1974	25.7	4	0.083333	-0.91024
7	1996	25.7	5	0.104167	-0.81614
8	1970	26.2	6	0.125	-0.7321
9	1995	26.2	7	0.145833	-0.65508
10	1984	26.7	8	0.166667	-0.5832
11	1961	27.3	9	0.1875	-0.5152
12	1965	27.3	10	0.208333	-0.45019
13	1973	27.3	11	0.229167	-0.38751
14	1977	27.3	12	0.25	-0.32663
15	1991	27.3	13	0.270833	-0.26716
16	1956	27.8	14	0.291667	-0.20876
35	1985	30.3	33	0.6875	0.981647
36	1987	30.3	34	0.708333	1.064673
37	1990	30.3	35	0.729167	1.152479
38	1971	30.9	36	0.75	1.245899
39	1983	30.9	37	0.770833	1.345985
40	1994	30.9	38	0.791667	1.454081
41	1952	31.4	39	0.8125	1.571953
42	1962	31.9	40	0.833333	1.701983
43	1972	31.9	41	0.854167	1.847511
44	1968	32.4	42	0.875	2.013419
45	1975	32.9	43	0.895833	2.207267
46	1953	33.4	44	0.916667	2.441716
47	1988	34	45	0.9375	2.740493
48	1959	36.5	46	0.958333	3.156849
49	1998	42.2	47	0.979167	3.860693

m

$$p = \frac{m}{N+1}$$

$$y = -ln[-ln(p)]$$

N

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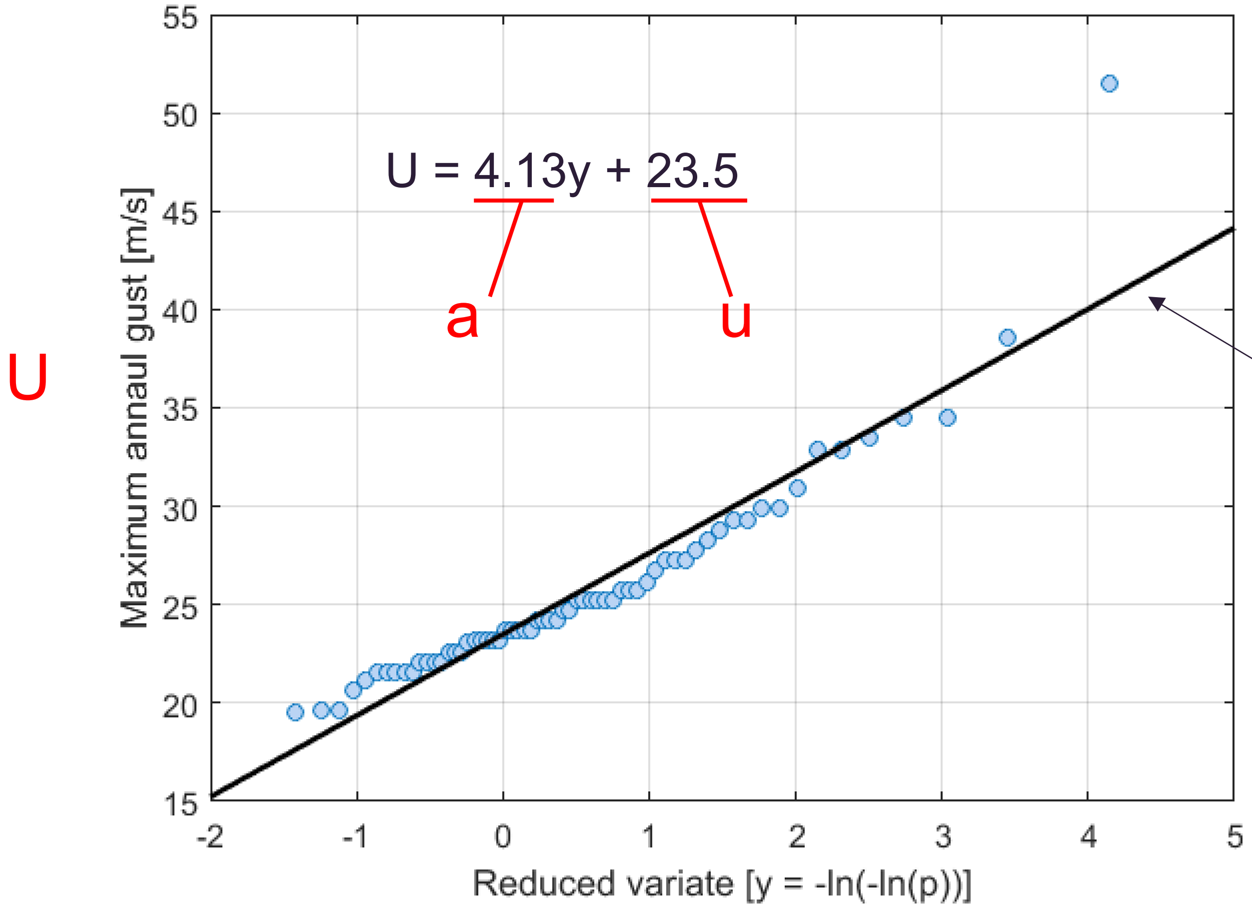
Example



How this line is fit will influence results

Example

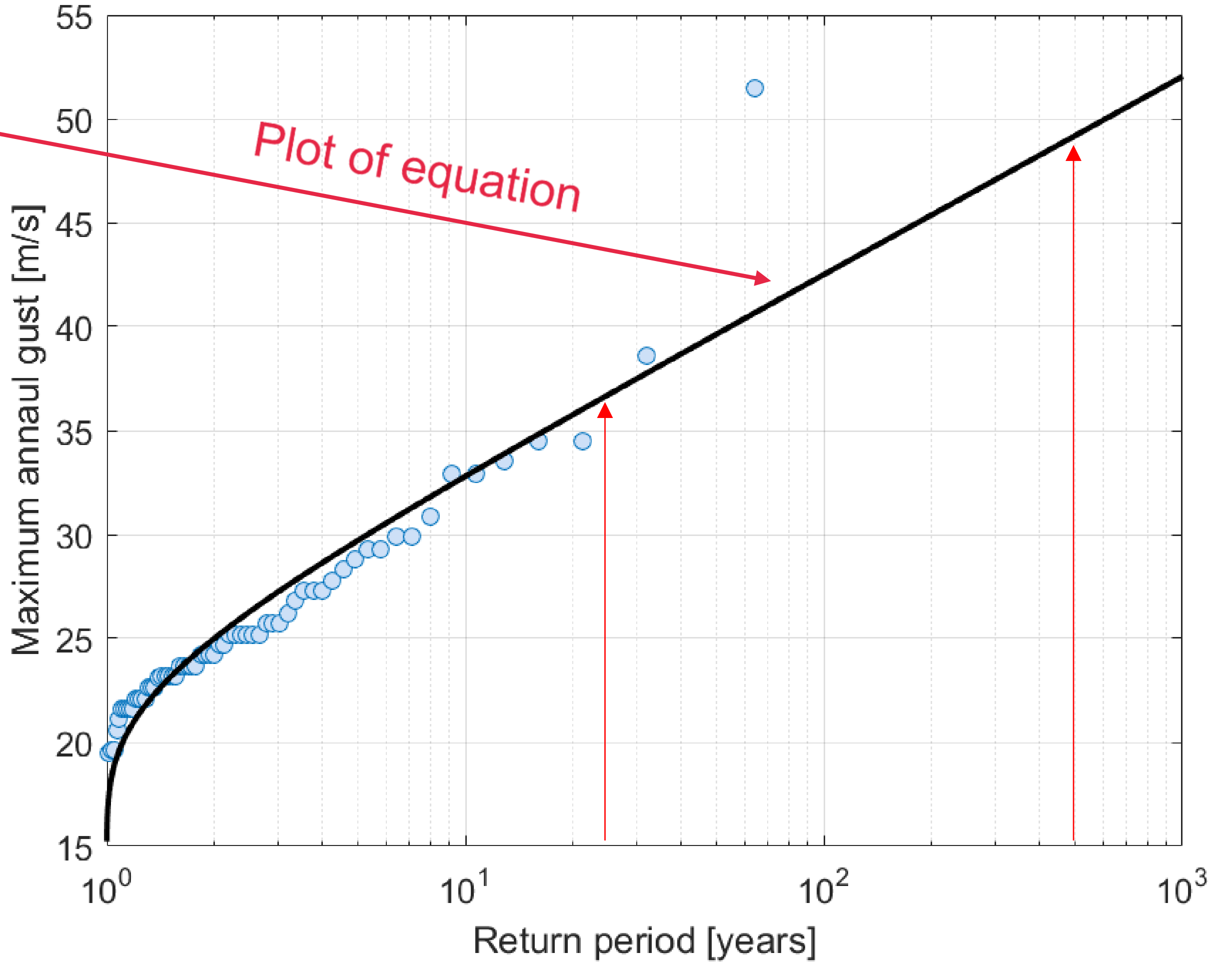
$$U = u + a\left\{-\ln\left[-\ln\left(1-\frac{1}{R}\right)\right]\right\}$$

- Using *u* and *a, U* can be calculated for any R you want.

- Data points can be plot by estimating the *R* for each point using:

$$R = \frac{1}{1-p}$$

ANSWER:
U_{25} = 36.7 m/s [*39 m/s*]
U_{500} = 49.2 m/s [*57 m/s*]



Alternate distributions

Other approaches exist for fitting (epoch) extrema

- Modified Gumbel methods (e.g. Gringorten)
- Method of moments
- Generalised penultimate distribution

Other, non-epoch, methods also exist.

- Peaks (excess) over threshold
- Method of Independent Storms
- Others…

Alternate distributions

Modifications to Gumbel's method

- ***Gringorten's Method***
  - Traditional Gumbel method is biased for high values of *p* (i.e. high return periods)
  - Gringorten (1953) proposed an unbiased modification:

$$p = \frac{(m - 0.44)}{(N + 0.12)} \quad \text{replaces} \quad p = \frac{m}{N + 1}$$

Brisbane Airport

R [Years]	Gumbel [m/s]	Gringorten [m/s]
10	32.9	32.6
100	42.7	41.9
1000	52.3	51.1

Alternate distributions

Modifications to Gumbel's method

- *Method of Moments*
  - Simple method only reliant on basic statistics (i.e. no direct fitting to data).

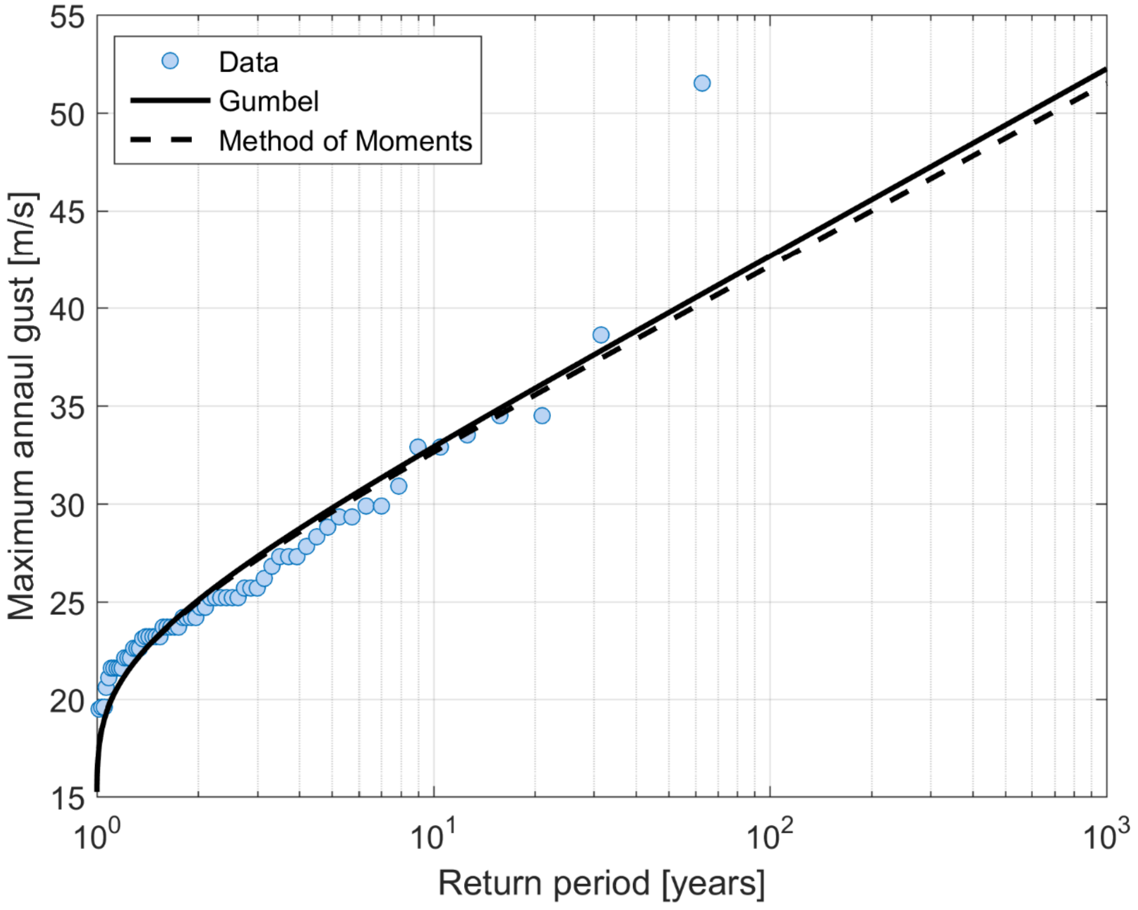
$$U = u + a\left\{-\ln\left[-\ln\left(1 - \frac{1}{R}\right)\right]\right\} = u + a\{y\}$$

Estimate these variables using data moments

mean $\mu = u + 0.5772a$

Standard deviation $\sigma = \left(\frac{\pi}{\sqrt{6}}\right)a$

$$a = \left(\frac{\sqrt{6}}{\pi}\right)\sigma \qquad u = \mu - 0.5772a$$



R [Years]	Gumbel [m/s]	MoM [m/s]
10	32.9	32.6
100	42.7	42.2
1000	52.3	51.5

Alternate distributions

Modifications to Gumbel's method

- *General Penultimate Distribution*
  - Cook and Harris (2004) proposed modifications to Gumbel based on parent distribution (assumed to be Weibull)

$$F(U) = e^{-\left[e^{-\left(\frac{U-u}{a}\right)}\right]}$$

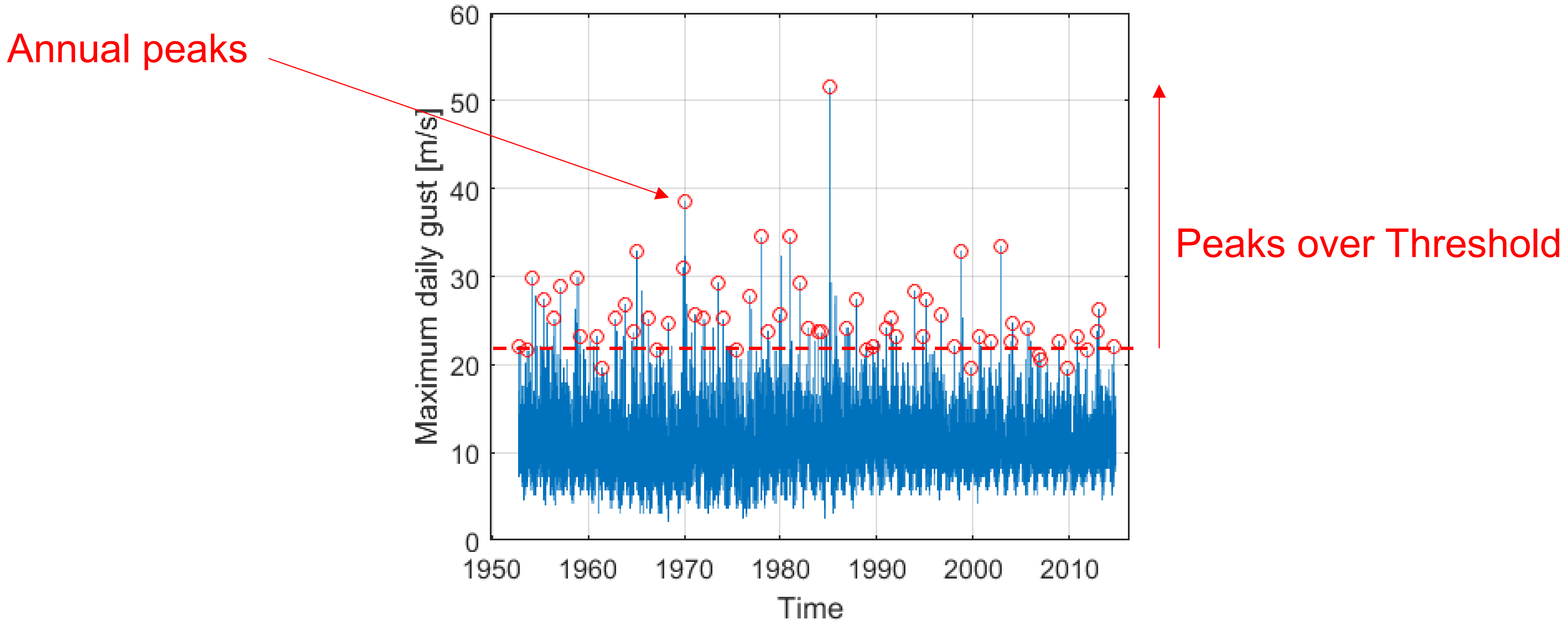
Gumbel

$$F(U) = e^{-\left[e^{-\left(\frac{U^{w}-u^{w}}{a^{w}}\right)}\right]}$$

Weibull exponent

  - Generates distributions that look like GEV Type III, but are not strictly bounded in the same way.

Alternate Methods



Brisbane Airport

Alternate distributions

Peaks over threshold

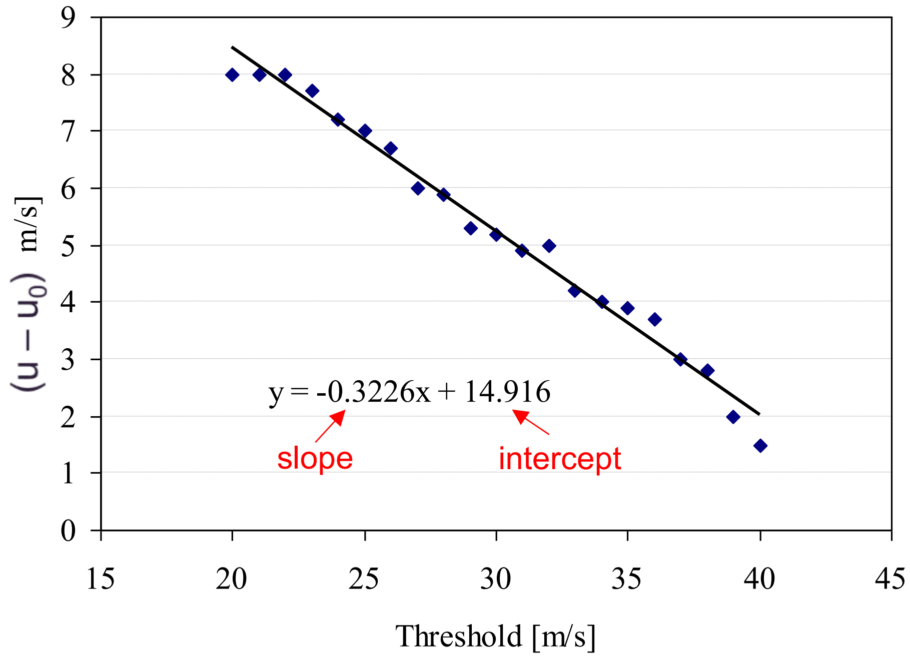
- Often uses Generalise Pareto Distribution
- Advantage: Can include more than one event per year
- Disadvantage: Minimum threshold is subjective, choice can influence results

$$U_R = u_0 + \sigma\left[1-(\lambda R)^{-k}\right]/k$$

Mean crossings per year

$$slope = \frac{-k}{(1+k)}$$

$$\text{int}\,ercept = \frac{\sigma}{(1+k)}$$



Peaks over threshold

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Method (graphical)

1. Specify a series of thresholds (e.g. 20, 21, 22…..m/s)
2. Calculate the number of positive crossings of the lowest threshold per year, λ
3. Calculate the difference between the crossing value and the max of each cross – average for each threshold
4. Plot mean excess against threshold level
5. Calculate the scale and shape factors σ, k.

Numerical fitting methods

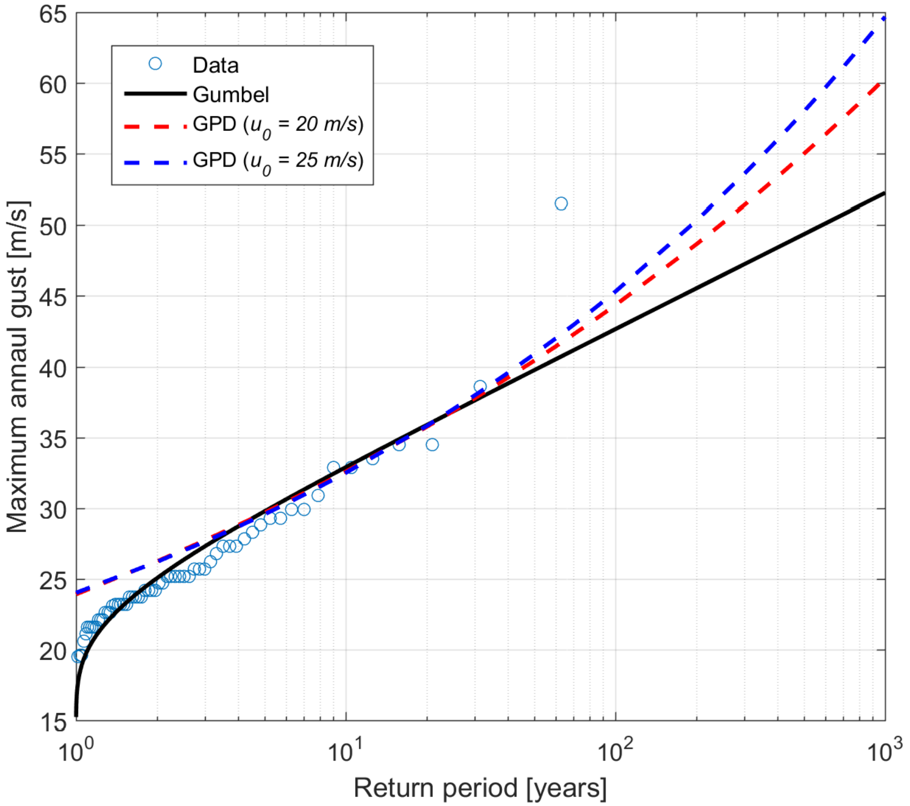
- E.g. MATLAB – Parms = gpfit($u_{u>u0} - u_0$)

  [-k, σ]

- Note: when plotting data, $R = 1/ [\lambda(1-p)]$

Exercise

- Using the Brisbane Airport data, apply the Peaks over Threshold method to determine 10 year, 100 year and 1000 year return period gusts. Compare these with standard Gumbel estimates.



R [Years]	Gumbel [m/s]	GPD (u_0 = 20) [m/s]
10	32.9	32.6
100	42.7	44.4
1000	52.3	60.3

Alternate distributions

Method of Independent Storms

- Similar to PoT, but uses storm information rather than thresholds.
- Advantage: Can explicitly include all storm events and separation by storm type becomes clearer.
- Disadvantage: Need very detailed records of not only wind, but any other parameter that is needed to identify storms.
- Often coupled with Generalised Pareto Distribution.

summary

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Reasons for undertaking a wind climate analysis

- Determining design wind speeds
- Fatigue estimation
- Natural ventilation, ESD
- Wind power assessment.

3 asymptotic forms of the Generalised Extreme Value (and Generalised Pareto) Distribution – each with different physical justification.

Several analysis techniques shown.

Contact

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